Step 0: Initial Document Check

- **0.1:** The document is an RCC structural drawing of "FOUNDATIONS" only.
- **0.2:** The site location is "D.K.Z.P HIGHER PRIMARY SCHOOL AT BOLLYARU".
- **0.3:** All compliance checks will be based only on IS 456:2000 and SP 34.

Step 1: Locate the "NOTES" Section

- **Status:** Compliant (The "NOTES" section is clearly present).

Step 2 & 3: Extract and Verify Design Parameters from "NOTES"

Criteria	Extracted Value	Compliance Check	Status
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1. Grade of Concrete	M25 for all R.C.C works.	M25 is a suitable grade for general RCC works. Environmental exposure conditions are not explicitly mentioned, but M25 is generally acceptable for moderate conditions.	Compliant
2. Reinforcement Bars	HYSD TMT bars of Grade Fe 500 conforming to IS 1786-1985. Vertical reinforcement and ties are specified in column schedules (e.g., 4Y16(d)+2Y12, Y8@200 for C1).	Fe 500 is a standard and compliant grade. Shear reinforcement (ties) are mentioned.	Compliant
3. Lap Length	50 times dia of the bar, to be staggered such that not more than 50% of the bars are lapped at a section.	"50 times dia of the bar" (50d) is compliant with SP 34. Staggering is also a good practice.	Compliant
4. Clear Cover	Footing/Wall: 50 mm, Columns: 40 mm, Slab: 20 mm, Beam: 25 mm.	Footing cover (50mm) is compliant. Column cover (40mm) is compliant. Slab (20mm) and Beam (25mm) covers are compliant. PCC is shown in the typical footing section, which is appropriate for a 50mm cover.	Compliant
5. Development Length (Ld)	50 times the dia of the bar.	"50 times the dia of the bar" (50d) is a common and generally compliant value.	Compliant
6. Safe Bearing Capacity (SBC) of soil	21 T/m2. PCC is shown in the typical footing section.	SBC is mentioned. PCC is shown in the typical footing section.	Compliant
7. Seismic Zone and Wind Load	Not mentioned.	This information is mandatory for structural design in India.	Missing Information
8. Building Limitations	Structure is designed for Ground + 1 Stories only.	Building limitations are mentioned.	Compliant
9. Structure's Purpose	D.K.Z.P HIGHER PRIMARY SCHOOL AT BOLLYARU.	The purpose is clearly stated.	Compliant

10. Floor Heights	Plinth Beam, First Floor, Terrace Lvl. (Specific heights not explicitly given in notes, but implied by column schedules).	Specific floor heights are not explicitly listed in the "NOTES" section, but the column schedules indicate "Foundation to First Floor" and "First Floor to Terrace Lvl.", implying floor levels. However, exact heights are not stated.	Missing Information
11. Schedule of Footings	A "SCHEDULE OF FOOTINGS" table is present, listing F1, F2, F3, F4 with Length, Width, Depth, and Bottom Reinforcement. The layout drawing uses these footing marks.	The drawing is consistent with the "SCHEDULE OF FOOTINGS" table.	Compliant
12. Footing Type	Isolated footings (F1, F2, F3, F4) are shown in the "FOUNDATION LAYOUT".	Isolated footings are used for a Ground + 1 storey school building, which is a low-rise structure. This is appropriate.	Compliant
13. Reinforcement in High-Rise Buildings	Not applicable (building is G+1 storey).	The building is not high-rise.	Not Applicable
14. Raft Foundation Reinforcement	Not applicable (isolated footings are used).	Raft foundation is not used.	Not Applicable
15. Lift Design	Not mentioned, no lift pit shown.	No lift design is present.	Not Applicable
16. Soil Improvement	Not mentioned.	No details about soil improvement methods are provided.	Missing Information
17. Column Ties	Column ties are shown in the column schedules (e.g., Y8@200, Y8@400). They appear continuous in the typical column sections.	Ties are shown and appear continuous in the typical sections.	Compliant
18. Plan of Ties	A separate plan for ties is not explicitly present; tie details are part of the column schedules.	A separate plan for ties is not provided, but details are in column schedules.	Missing Information
19. Outer Ties Check	Column schedules specify tie diameters and spacing (e.g., Y8@200, Y8@400). Percentage of steel in columns is not explicitly stated in notes but can be calculated from column schedules. Footing reinforcement is given (e.g., Y10@175).	Tie specifications are present. Percentage of steel in columns and footings is not explicitly stated as a general note, but reinforcement details are provided for calculation.	Cannot Verify
20. Cross-Section Area	Not mentioned.	It is not specified whether gross or effective cross-section area is used.	Missing Information
21. Steel Curtailment	Not mentioned.	Curtailment is not explicitly detailed in the notes. For a G+1 building, it might not be as critical as for high-rise, but general practice notes would be beneficial.	Missing Information

22. Maximum Steel Percentage in Columns	Not mentioned.	The maximum percentage of steel in columns is not stated in the notes.	Missing Information

Step 5: Report Missing or Wrong Information

Site Location: While a site name is given, specific geographical coordinates or a more precise address for environmental exposure assessment are missing.

Seismic Zone and Wind Load: This crucial information for structural design is entirely absent from the notes.

Floor Heights: While floor levels are implied, the exact height between floors is not explicitly stated in the notes.

Soil Improvement: No details regarding any soil improvement methods used at the site are provided.

Plan of Ties: A dedicated plan for column ties is not present; details are embedded within column schedules.

Cross-Section Area: It is not specified whether gross or effective cross-section area is used for design calculations.

Steel Curtailment: Details regarding steel curtailment are not provided.

Maximum Steel Percentage in Columns: The maximum percentage of steel allowed in columns is not stated in the notes.

Summary of Compliance

• **Total Criteria Evaluated:** 22

• **Compliant Items:** 8

• **Non-Compliant Items:** 0

• **Missing Information Items:** 8

• **Cannot Verify:** 1

• **Not Applicable:** 3

• **Overall Verdict:** The drawing has significant missing information, particularly regarding seismic and wind loads, which are critical for structural design compliance in India. While many basic parameters are compliant, the absence of key design inputs makes a full compliance assessment difficult and raises concerns about the completeness of the design documentation.